COMP30022 IT Capstone Project

Semester 2, 2025

Ethics and Professionalism

Topics for today's lecture

- Professionalism
- Codes of ethics
- Stories
- Lessons to be learned

Traffic cameras

- My first time as an expert witness
- Redflex consulting

Is an IT worker a professional?

- What is professional behaviour?
- What if you don’t exhibit it?
- Analogy to doctor and accountant
- Qn: Does it matter?

Codes of Ethics

- What is their purpose?
  - On Canvas for you to look at …
- Australian Computer Society Code
- Engineers Australia
- Software Engineering

ACS CODE OF ETHICS

The ACS Code of Ethics is part of the ACS Constitution. As an ACS member you must uphold and advance the honour, dignity and effectiveness of being a professional. This entails, in addition to being a good citizen and acting within the law, your adherence to the following Society values:

1 The Primacy of the Public Interest
You will place the interests of the public above those of personal, business or sectional interests.

2 The Enhancement of Quality of Life
You will strive to enhance the quality of life of those affected by your work.

3 Honesty
You will be honest in your representation of skills, knowledge, services and products.

4 Competence
You will work competently and diligently for your stakeholders.

5 Professional Development
You will enhance your own professional development, and that of your colleagues and staff.

6 Professionalism
You will enhance the integrity of the Society and the respect of its members for each other.

This Code of Ethics applies to all ACS members regardless of their role or specific area of expertise in the ICT industry.

ACS
Code

Compliance to the law

- Downloading music
  - Most students do it and are not concerned
- Mp3s4free case
- Kazaa case
- What should be built
- How much control do the companies have?

Engineers Australia Code of Ethics

1 Demonstrate integrity

1.1 Act on the basis of a well-informed conscience

1.2 Be honest and trustworthy

1.3 Respect the dignity of all persons.

2 Practise competently

2.1 Maintain and develop knowledge and skills

2.2 Represent areas of competence objectively

2.3 Act on the basis of adequate knowledge.

3 Exercise leadership

3.1 Uphold the reputation and trustworthiness of the practice of engineering

3.2 Support and encourage diversity

3.3 Make reasonable efforts to communicate honestly and effectively to all stakeholders, taking into account the reliance of others on engineering expertise.

4 Promote sustainability

4.1 Engage responsibly with the community and other stakeholders

4.2 Practise engineering to foster the health, safety and wellbeing of the community and the environment

4.3 Balance the needs of the present with the needs of future generations.

Four famous examples of problematic behaviour

- Volkswagen emissions scandal
  - https://en.wikipedia.org/wiki/Volkswagen_emissions_scandal
- Facebook scandals
  - https://www.businessinsider.com/mark-zuckerberg-scandals-last-decade-while-running-facebook-2019-12
- Therac-25
  - https://www.codereliant.io/p/the-most-tragic-bug
- Pentium FDIV bug

Lessons to be learned

- Aim to be professional
- Care about ethics
  - Choose your workplace accordingly
- Remember you were taught about ethics and security 😀